

$$1. R_{\Sigma_1} = \{a, b\}$$

$$R_{\Sigma_2} = \{b\}$$

$$R_{\Sigma_3} = \{0, 1\}$$

22/09

* Complementation:

$$\Sigma = \{a, b\}$$

$$R = \{w \mid \text{odd b's and even a's} \\ \text{a's followed by b's} \\ \text{of length at least 1} \\ \text{such that there is} \\ \text{at least one occurrence} \\ \text{of each}\}$$

$$= \{ab, abb, aab, \dots\}$$

$$\Sigma^* = \{\dots, ab, aa, ba, bb, \dots\}$$

$$\begin{array}{l} abb \\ aab \end{array} \quad \Sigma^* - R = R^c \\ = \{\epsilon, a, b, aa, ba, bb, \dots\}$$

$$Q. \Sigma = \{a\}$$

$$R_1 = \{w \mid \text{string of a's of odd length}\} \\ (\text{Regular set})$$

$$r_1 = (\text{Regular expression}) \\ = a(aa)^*$$

$$\Sigma_1^* \setminus R_1 \neq \emptyset$$

$$R_1 = \{a, aaa, aaaaa, \dots\}$$

$$\Sigma_1^* \setminus R_1 = R_1^c = \{\epsilon, aa, aaaa, \dots\}$$

a's followed by b's such that

$$Q. R_3 = \{w \mid \text{minimum occurrence of a being 1}\}, \Sigma = \{a, b\}$$

$$= \{a, aa, ab, aaa, abb, aab, \dots\} = a^+ \cup ab a(a^*)(b^*)$$

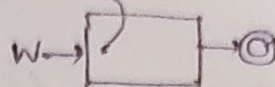
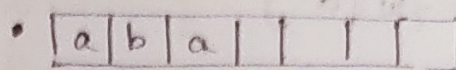
$$\therefore R_1 \subseteq R_3 \{a, b\}$$

• In transition graph the transition happens from state to another.

• In tape it is right.

• For two way automata:

$$\delta: Q \times \Sigma \times \{L, R\} \rightarrow Q$$



$$\begin{array}{l} a(aa)^* = a\{\epsilon, aa, aaaa, \dots\} \\ = \{a, aaa, aaaaa, \dots\} \end{array}$$

• $\|w\|$ = Length of String.

$|R|$ = Cardinality of set

• Equivalence between:

Regular set $\cong$ Finite automata $\cong$ Transition graph.

• Always mention alphabet.

• Regular set is a subset of Universal set

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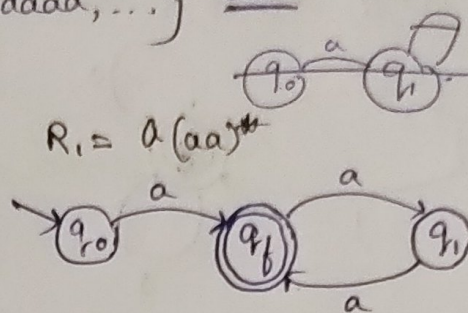
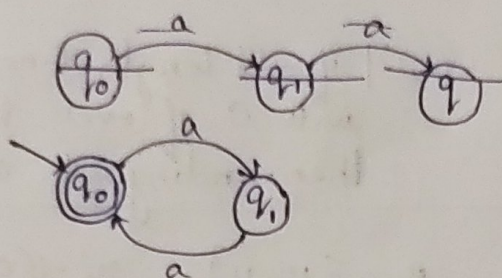
Q. $r = a(ab)^+ = \text{NOT A REGULAR EXPRESSION}$

Q. $\Sigma' = \{a, bb\}, \Sigma'^*$

$w = bba$; $\|w\| = 2$; $babb \notin \Sigma'^*$

Transition Graphs

1. $\Sigma'^* \setminus R_1 = R_1^c = (aa)^* = \{\epsilon, aa, aaaa, \dots\}$ aa^*



2. $r = a(ab)^+$ is equivalent to $r' = a(a.b)^*ab$ but not equal because the first one is ~~an~~ a regular expression.

3. $a^+ \equiv a.a^*$ same for these two.

Configuration: $\Sigma^* \times Q \rightsquigarrow$

initial configuration: $(w, q_0) \vdash (w', q_1)$

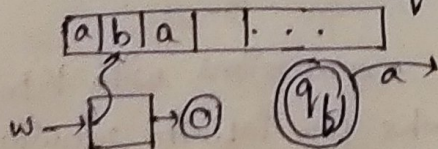
$\vdash (\epsilon, q_f)$, if q_f is a final state then only*

How each symbol ~~is read~~ in a regular expression is read past.

$w = aw'$

$\delta(q_0, a) = q_1$

$(w, q_0) \vdash (w', q_1)$
 $\vdash (w'', q_f)$



* we can say that the input string w has been accepted by the automata, otherwise it would not be a final configuration and we can conclude that the string is not accepted.

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1. $\Sigma = \{0, 1\}$

$\pi_1 = \epsilon$

$R_\Sigma = \{\epsilon\}$

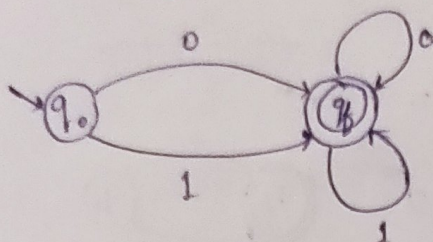
$R^c = \{0, 1, 00, 01, \dots\}$

$\pi_1 = 1(0+1)$

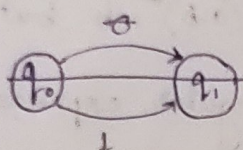
$R_1 = \{10, 11\}$

$R_1^c = \{\epsilon, 0, 1, \dots\}$
 $= \Sigma^* \setminus R_1$

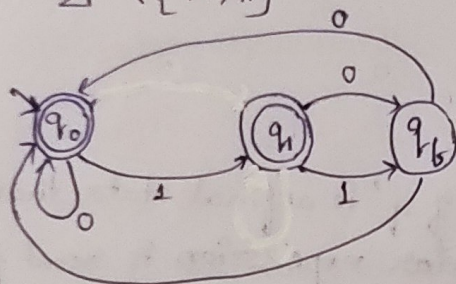
$R^c = \Sigma^* \setminus R_\Sigma = (0+1)^*(0+1)$



$R_1^c = \{\epsilon, 0, 1, 00, 01, \dots\}$



$= \Sigma^* \setminus \{10, 11\}$



δ	0	1
q_0		

$\pi_2 = (10+11)^* 1$

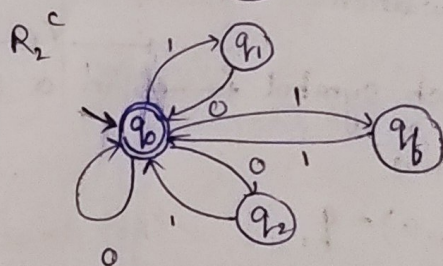
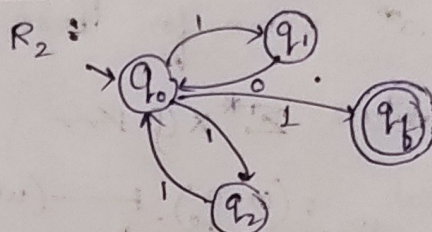
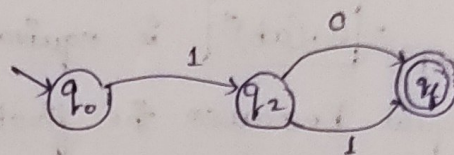
$R_2 = \{1, 101, 111, 1011, 1101, 10101, 11111, \dots\}$

$R_2^c = \{\epsilon, 0, 10, 11, 01, 00, 000, 001, \dots\}$

$R_2 = \{w \mid \text{odd length ending with } 1\}$

$R_2^c = \{w' \mid \text{if odd length then ending with } 0, \text{ if even length then ending with } 0 \text{ or } 1\}$

$R_1 = \{10, 11\}$ $\pi_1 = 1(0+1)$



$L_0 \supset L_1 \supset L_2 \supset L_3$ Pumping Lemma - Regular Set.

States don't get added in automata, non-final states become final.

$m = (Q, \Sigma, \delta, q_0, F)$ for R^c

$m' = (Q, \Sigma, \delta', q_0, Q \setminus F)$ for regular expressions with Kleene closure, works for infinite regular sets.

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* Intersection:-

$$R_1 \cap R_2 = \Sigma^* \setminus ((\Sigma^* \setminus R_1) \cup (\Sigma^* \setminus R_2))$$

$$R_1 \cap R_2 = \Sigma^* - (R_1^c \cup R_2^c)$$

* Pumping Lemma: Let, L be a regular set/language and let there be an integer $n \geq 1$ such that an arbitrary string $w \in L$ with length $\|w\| \geq n$ then, if

$$w = xyz, \quad y \neq \epsilon$$

and $\|xy\| \leq n$ are true then,

$$xy^i z \in L, \quad i \geq 1.$$

• $L(M)$ = The regular set/language accepted by the finite automata.

• $L(R)$ = The regular expression accepted by a language.

• If there is a regular set then there is a finite automata (deterministic or non-deterministic)

↳ Proof: $L = L(M) \quad M = (Q, \Sigma, \delta, q_0, f)$

$$i. |Q| = n$$

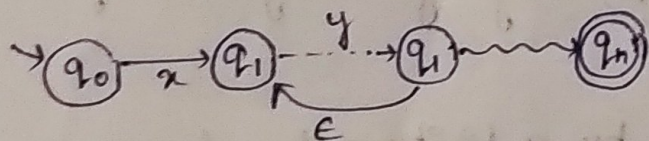
• Now, $\|xy\| \leq n$

if $\|xy\| = n$

$$q_0 \rightarrow q_1 \rightarrow \dots \rightarrow q_i \rightarrow \dots \rightarrow q_n$$

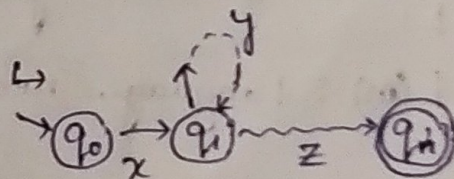
↳ no. of states = $n+1$

↳ 1 state is repeated.



• For, $\|xy\| < n$

• Proof look it up.



$$= xy^i z, \quad i \geq 1$$

There exists a DFA for equivalent for every NFA.

↳ For every NFA there exists an equivalent DFA.

The two finite automata are said to be equivalent if the two accept the same state.

Minimize automata

Finite automata for complementary set may be non-deterministic.

29/09/2023.

Let $R = 1.(0+1)$ where $\Sigma' = \{0, 1\}$

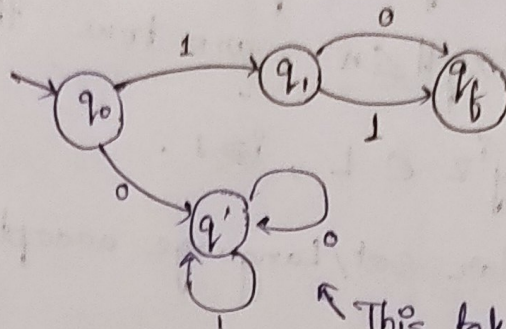
$$\Sigma' \subset \Sigma$$

$$\therefore R = 10 + 11^*$$

$$\Sigma^{*} = \{w_0, w_1 \mid w \in \Sigma^{*}\}$$

w_0 = strings starting with 0

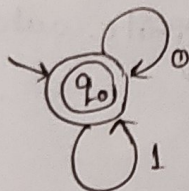
w_1 = strings starting with 1



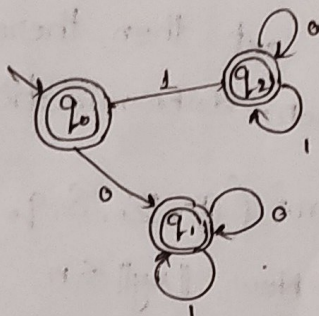
↑ This takes care of w_0 .

Finite, $M' = (Q', \Sigma, \delta', q_0, Q' \setminus F)$

Σ^{*}



$\equiv$



$R \begin{cases} \text{finite} \\ \text{infinite} \end{cases}$

$\equiv$ Improve such that $\{10, 11\}$ is not accepted.

$R_2 = \{w \mid w \in L(\pi_2)\}$ where $\pi_2 = (10+01)^*$. † Give expression of R_2^c .

$L(T)$: Transition graph is given, Language recognised by Transition graph.

$L(M)$: Language recognised by a finite automata.

incl.

ie sets.

say $w' = 0w''$

$$\delta(q_0, 0) = q_1$$

$$(w', q_0) \vdash (w'', q_1)$$

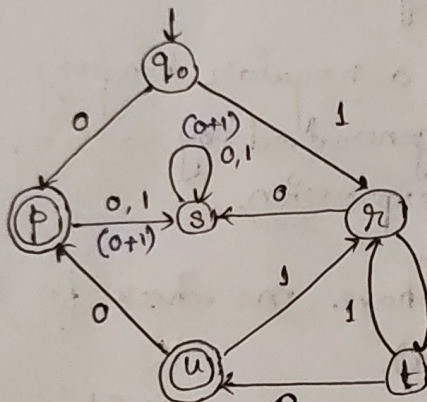
$$\vdash (e, q_f)$$

q_f one of the final states $\in F$

This string has been accepted by the automata, i.e., a valid string of $L(M)$.

If $\vdash (e, q_j)$, where, $q_j \notin F$, then string is not accepted by automata.

AE
OCCAM
↓
Razor



$$\Sigma = \{0, 1\}$$

Three types of nodes:

i. Initial state (q_0)

ii. Final states $(p), (u)$

iii. Normal states $(s), (q_1), (t)$

Check if $w \in L(M)$
or $e \in L(T)$

It is not a non-deterministic automata.

$$\text{e.g. } \delta(p, 0) = s$$

$$\delta(p, 1) = s$$

Non-determinism is:

$$\delta(q_i, 0) = q_j$$

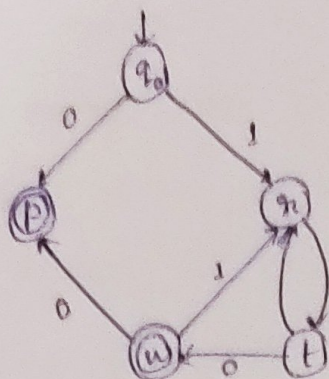
$$\delta(q_i, 1) = q_k$$

Example is conditional statements like if... else, switch case, etc.

S. Motive
Feature
→ Mass
→ On de
→ Data
New
ing par
center
on b/w
Data co

in close

2023



$$0 + 11(u)^* (0 + 00)$$

- If two automata recognise/accept the same language then they are called equivalent.

• Equivalences

$$\begin{matrix} m \\ L \\ L' \end{matrix} \quad \begin{matrix} m' \end{matrix}$$

- Two languages L and L' are said to be equivalent if they are accepted by the same automata.

- $L(G)$: a regular language generated by a regular grammar.

• Definitions $\delta^*(q_0, w)$

$$\delta^*: Q \times \Sigma^* \rightarrow \mathcal{P}(Q)$$

$$\text{e.g. } \delta^*(q_0, 110) = \{r, t, u\}$$

$$\{r, t, u\} \subseteq Q$$

$$\{r, t, u\} \in \mathcal{P}(Q)$$

- $L(R)$: a regular language generated by a regular expression.

- If we have m check for $L(G)$ or $L(R)$.

- It gives us a sequence of states.

$$\delta^*(u, 11) = \{r, t\}$$

- The last element in a sequence might be a final state.